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Granite-Embedding-30m-English (revision r1.1)

Model Summary: Granite-Embedding-30m-English is a 30M parameter dense bi-encoder embedding model from the Granite Embeddings suite that can be used to generate high quality text embeddings. This model produces embedding vectors of size 384 and is trained using a combination of open source relevance-pair datasets with permissive, enterprise-friendly license, and IBM collected and generated datasets. While maintaining competitive scores on academic benchmarks such as BEIR, this model also performs well on many enterprise use cases. This model is developed using retrieval oriented pre-training, contrastive fine-tuning, knowledge distillation and model merging for improved performance.

Granite-embedding-30m-r1.1 was specifically designed to support multi-turn information retrieval and is designed to handle contextual document retrieval in multi-turn conversational information retrieval. Granite-embedding-30m-r1.1 was trained on data tailored for multi-turn conversational information retrieval and uses multi-teacher distillation over granite-embedding-30m-english (https://huggingface.co/ibm-granite/granite-embedding-30m-english)

- **Developers:** Granite Embedding Team, IBM
- **GitHub Repository:** ibm-granite/granite-embedding-models
- **Website**: Granite Docs

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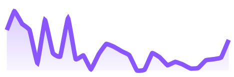

Safetensors

Model size	30.3M params
Tensor type	BF16

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`ibm-granite/granite-embedding-30m-english` is supported by the following Inference Providers:

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mteb/leaderboard

aploetz/granite-embeddings

ipepe/nomic-embeddings

niloydebbarma/allycat

Prathamesh1420/maintenance_chat...

- **Paper:** Technical Report
- **Release Date**: August 29, 2025 (granite-embedding-30m-english-r1.1)
- **License:** Apache 2.0

Supported Languages: English.

Intended use: The model is designed to produce fixed length vector representations for a given text, which can be used for text similarity, retrieval, and search applications.

Usage with Sentence Transformers: The model is compatible with SentenceTransformer library and is very easy to use:

First, install the sentence transformers library

```
pip install sentence_transformers
```

The model can then be used to encode pairs of text and find the similarity between their representations.

Granite-Embedding-30m-English

```
from sentence_transformers import SentenceTra

model_path = "ibm-granite/granite-embedding-3
# Load the Sentence Transformer model
model = SentenceTransformer(model_path)

input_queries = [
    ' Who made the song My achy breaky heart'
    'summit define'
    ]

input_passages = [
    "Achy Breaky Heart is a country song wri
    "Definition of summit for English Langua
    ]

# encode queries and passages
query_embeddings = model.encode(input_querie
passage_embeddings = model.encode(input_pass

# calculate cosine similarity
print(util.cos_sim(query_embeddings, passage_
```

Granite-Embedding-30m-r1.1

maxpar1/leaderboard

kiedistv/leaderboard

malyvena/leaderboard + 7 Spaces

Collection including `ibm-granite/…`

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Specifically to encode with granite-embedding-30m-r1.1 the entire conversation, ending with the last user query, should be provided as the input, with the conversation instances arranged in reverse chronological order: first the last user query, then the preceding agent response, followed by the previous user query. Example,

Conversation: user: agent: user: agent: user: agent: user:

Conversation in input query format: [SEP]agent: ||user:||agent: ||user:||agent: ||user:

```
from sentence_transformers import SentenceTra

model_path = "ibm-granite/granite-embedding-3
# Load the Sentence Transformer model
model = SentenceTransformer(model_path, revis

input_queries = [
    "Which team has won the most Super Bowls

    "How many teams are in the NFL playoffs?
    ]

input_passages = [
    "Super Bowl\nThe Pittsburgh Steelers have

    "NFL playoffs \n The 32 - team National F
    ]

# encode queries and passages
query_embeddings = model.encode(input_queries
passage_embeddings = model.encode(input_passa

# calculate cosine similarity
print(util.cos_sim(query_embeddings, passage_
```

Usage with Huggingface Transformers: This is a simple example of how to use the Granite-Embedding-30m-English model with the Transformers library and PyTorch.

First, install the required libraries

```
pip install transformers torch
```

The model can then be used to encode pairs of text

Granite-Embedding-30m-English

```
import torch
from transformers import AutoModel, AutoToke

model_path = "ibm-granite/granite-embedding-

# Load the model and tokenizer
model = AutoModel.from_pretrained(model_path
tokenizer = AutoTokenizer.from_pretrained(mo
model.eval()

input_queries = [
    ' Who made the song My achy breaky heart
    'summit define'
    ]

# tokenize inputs
tokenized_queries = tokenizer(input_queries,

# encode queries
with torch.no_grad():
    # Queries
    model_output = model(**tokenized_queries
    # Perform pooling. granite-embedding-30m
    query_embeddings = model_output[0][:, 0]

# normalize the embeddings
query_embeddings = torch.nn.functional.norma
```

Granite-Embedding-30m-r1.1

```
import torch
from transformers import AutoModel, AutoToke

model_path = "ibm-granite/granite-embedding-

# Load the model and tokenizer
model = AutoModel.from_pretrained(model_path
tokenizer = AutoTokenizer.from_pretrained(mo
model.eval()

input_queries = [
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    "How many teams are in the NFL playoffs?
    ]

# tokenize inputs
tokenized_queries = tokenizer(input_queries,

# encode queries
with torch.no_grad():
    # Queries
    model_output = model(**tokenized_queries
    # Perform pooling. granite-embedding-30m
```

```
    query_embeddings = model_output[0][:, 0]

# normalize the embeddings
query_embeddings = torch.nn.functional.norma
```

Evaluation:

Granite-Embedding-30M-English model is twice as fast as other models with similar embedding dimensions, while maintaining competitive performance. The performance of the Granite-Embedding-30M-English model on MTEB Retrieval (i.e., BEIR) and code retrieval (CoIR) benchmarks is reported below.

Model	Paramters (M)	Embedding Dimension	MTEB Retrieval (15)	CoI (10
granite-embedding-30m-english	30	384	49.1	47.(

granite-embedding-30m-r1.1 revision maintains the fast speed of granite-embedding-30m-english while demontratring strong performance on multi-turn information retrieval benchmarks. The performance of the granite-embedding-30M-r1.1 model on MTEB Retrieval (i.e., BEIR) and multi-turn information retrieval (MTRAG(https://github.com/IBM/mt-rag-benchmark), Multidoc2dial(https://github.com/IBM/multidoc2dial)) datasets is reported below.

Model	Parameters (M)	Embedding Dimension	MTEB Retrieval (15)	M R
granite-embedding-30m-english	30	384	49.1	4
granite-embedding-30m-english-r1.1	30	384	48.9	**5**
bge-small-en-v1.5	33	512	53.86	3

Model	Parameters (M)	Embedding Dimension	MTEB Retrieval (15)	M R
e5-small-v2	33	384	48.46	2

Model Architecture: granite-embedding-30m-English is based on an encoder-only RoBERTa like transformer architecture, trained internally at IBM Research. granite-embedding-30m-r1.1 shares the same architecture as granite-embedding-30m-English

Model	granite-embedding-30m-english	granite-embedding-125m-english	granite-embedding-107m-multilingua
Embedding size	**384**	768	384
Number of layers	**6**	12	6
Number of attention heads	**12**	12	12
Intermediate size	**1536**	3072	1536
Activation Function	**GeLU**	GeLU	GeLU
Vocabulary Size	**50265**	50265	250002
Max. Sequence Length	**512**	512	512
# Parameters	**30M**	125M	107M

Training Data: Overall, the training data consists of four key sources: (1) unsupervised title-body paired data scraped from the web, (2) publicly available paired with permissive, enterprise-friendly license, (3) IBM-internal paired data targeting specific technical domains, and (4) IBM-generated synthetic data. The data is listed below:

Dataset	Num. Pairs
SPECTER citation triplets	684,100
Stack Exchange Duplicate questions (titles)	304,525
Stack Exchange Duplicate questions (bodies)	250,519
Stack Exchange Duplicate questions (titles+bodies)	250,460
Natural Questions (NQ)	100,231
SQuAD2.0	87,599
PAQ (Question, Answer) pairs	64,371,441
Stack Exchange (Title, Answer) pairs	4,067,139
Stack Exchange (Title, Body) pairs	23,978,013
Stack Exchange (Title+Body, Answer) pairs	187,195
S2ORC Citation pairs (Titles)	52,603,982
S2ORC (Title, Abstract)	41,769,185
S2ORC (Citations, abstracts)	52,603,982
WikiAnswers Duplicate question pairs	77,427,422
SearchQA	582,261
HotpotQA	85,000
Fever	109,810
Arxiv	2,358,545
Wikipedia	20,745,403
PubMed	20,000,000
Miracl En Pairs	9,016
DBPedia Title-Body Pairs	4,635,922
Synthetic: Query-Wikipedia Passage	1,879,093
Synthetic: Fact Verification	9,888
IBM Internal Triples	40,290
IBM Internal Title-Body Pairs	1,524,586
MultiDoc2Dial Train (MultiTurn Conversation)	21,451
Synthetic IBM internal data	19,533

Notably, we do not use the popular MS-MARCO retrieval dataset in our training corpus due to its non-commercial license, while other open-source models train on this dataset due to its high quality.

Infrastructure: We train Granite Embedding Models using IBM's computing cluster, Cognitive Compute Cluster, which is outfitted with NVIDIA A100 80gb GPUs. This cluster provides a scalable and efficient infrastructure for training our models over multiple GPUs.

Ethical Considerations and Limitations: The data used to train the base language model was filtered to remove text containing hate, abuse, and profanity. Granite-embedding-30m-english and Granite-embedding-30m-r1.1 are trained only for English texts, and has a context length of 512 tokens (longer texts will be truncated to this size).

Resources

- ⭐️ Learn about the latest updates with Granite: https://www.ibm.com/granite
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